SMART RESTAURANT MANAGEMENT SYSTEM

User Roles and Permissions

Who uses the system and what each role may do

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PHASE	Phase 1: event-driven core | Phase 2: AI camera detection
PURPOSE	Defines every actor, their goals, permissions, restrictions and authentication method.

1. Overview

The system has six human roles and two automated actors. Access is controlled by role-based access control (RBAC). Every staff action is tied to a user and recorded in the audit log.

Role	Who they are	Login method	Primary device
Customer (Guest)	Diner at a table	None required: QR session. Optional phone OTP for receipts	Own phone
Waiter / Server	Floor staff	Staff ID + PIN	Phone or tablet
Kitchen Staff	Chef, cooks	Staff ID + PIN	Kitchen tablet / screen
Cleaner / Busser	Table cleaning staff	Staff ID + PIN	Phone
Manager	Shift or floor manager	Email + password (optional 2FA)	Tablet / PC / phone
Owner / Admin	Owner or system administrator	Email + password + 2FA	PC
System (Rules Engine)	Timers and scheduled jobs	Internal service account	Backend
AI Vision Service (Phase 2)	Camera detection service	Signed service token	Edge or cloud server

2. Role Profiles

2.1 Customer (Guest)

Can do

- Open the menu from the table QR; search and filter (veg, category, price).
- Customise items; add to the shared table cart; place orders (multiple rounds allowed).
- See live order status per item.
- Call waiter, request bill, report an issue.
- Pay online or choose cash; split the bill (Phase 1.5); leave feedback.

Restrictions

- Cannot see other tables or other sessions.
- Cannot cancel items once the kitchen has accepted them (must ask staff).
- Session expires after payment or inactivity timeout.

2.2 Waiter / Server

Can do

- See assigned tables and their live state.
- Acknowledge and resolve alerts (call waiter, bill request).
- Place or edit an order on behalf of a guest; seat a walk-in manually.
- Mark items as Served.
- Record cash payments; request manager approval for discounts/cancellations.

Restrictions

- Cannot change menu or prices.
- Cannot issue refunds or view revenue reports.
- Sees only their section unless the manager grants more.

2.3 Kitchen Staff

Can do

- See incoming order tickets in time order with table number and notes.

- Accept, start preparing, mark Ready.
- Mark an item out of stock (temporarily unavailable).

Restrictions

- Cannot see payment data.
- Cannot cancel a paid order.
- Cannot edit prices.

2.4 Cleaner / Busser

Can do

- See the cleaning queue: tables in Needs Cleaning state, oldest first.
- Start a cleaning task and mark it Done.
- Flag a table as Out of Service (damage, spill) with a note.

Restrictions

- Cannot view orders or payments.
- Cannot change any state except cleaning related ones.

2.5 Manager

Can do

- Live floor map with all tables, alerts, KPIs and timers.
- Override any table state with a reason (logged).
- Assign tables to waiters and tasks to cleaners.
- Approve discounts, cancellations and refunds up to a limit.
- Manage menu availability, staff shifts and view shift reports.
- Manage reservations and the waiting list.

Restrictions

- Cannot delete audit logs.
- Cannot change system-level or payment gateway settings (Owner only).

2.6 Owner / Admin

Can do

- Everything a manager can do, plus: create outlets, floors, tables and QR codes.
- Manage staff accounts and roles.
- Configure taxes, service charge, payment gateway keys, alert thresholds.
- Full reports, exports and audit log access.
- Configure camera integration (Phase 2).

Restrictions

- Actions on payment settings and refunds above limit require 2FA.

3. Permission Matrix

Y = allowed, - = not allowed, Own = only their own records or assigned tables, Appr = needs approval.

Capability	Guest	Waiter	Kitchen	Cleaner	Manager	Owner
Open menu / place order via QR	Y	-	-	-	-	-
Place / edit order for guest	-	Y	-	-	Y	Y
View kitchen tickets	-	Y	Y	-	Y	Y

Capability	Guest	Waiter	Kitchen	Cleaner	Manager	Owner
Update item status (accept, ready)	-	Served only	Y	-	Y	Y
View live floor map	-	Own section	-	-	Y	Y
Override table status	-	Limited	-	Cleaning only	Y	Y
Raise / view alerts	Call only	Y	-	-	Y	Y
Resolve alerts	-	Y	-	-	Y	Y
Cleaning tasks (start, done)	-	-	-	Y	Y	Y
Pay bill / record cash	Pay own bill	Cash	-	-	Y	Y
Discount / item cancel	-	Appr	-	-	Y	Y
Refund	-	-	-	-	Up to limit	Y
Edit menu and prices	-	-	Availability only	-	Y	Y
Manage tables and QR codes	-	-	-	-	-	Y
Manage staff and roles	-	-	-	-	Shifts only	Y
Reports	-	-	-	-	Y	Y (all outlets)
System / gateway settings	-	-	-	-	-	Y
View audit log	-	-	-	-	Read	Y

4. Authentication and Session Rules

- **Guests:** a short-lived session token is issued when the QR is scanned. It is bound to one table session and expires on payment, after the configured idle time, or when the manager closes the table.
- **Staff:** PIN login for floor roles (fast shift start); email and password for manager and owner. Failed attempts are rate limited and locked after repeated failure.
- **Tokens:** short-lived access token plus refresh token. Role and restaurant ID are embedded and verified on every request.
- **Multi-outlet:** a user belongs to one restaurant (outlet); owners can belong to several.
- **Audit:** logins, overrides, discounts, refunds, menu and settings changes are recorded with user, time and device.

5. Automated Actors

Actor	What it does	Allowed actions
System rules engine	Runs timers: idle session, order delay, long stay, unattended table	Raise alerts, create cleaning tasks, expire sessions
AI vision service (Phase 2)	Sends detection events (person seated, table cleared, queue size)	Create suggestions or set status per configured confidence rules; never edit payments